TOOLS: kmerseek 0.4 (3c353d4), hsdm17.k6.s1.lcfalse.extend-c0.18; query H7FWB6 (Flavobacterium frigoris), target database reviewed Swiss-Prot 2026_03
HYPOTHESIS: a cross-lineage hit should match a physicochemical pattern without matching residues
CONCLUSION: the only cross-lineage pair kmerseek reaches, 53 times across settings; both proteins are in UniProt's Ice-binding protein family, so they are homologs

```
hsdm17.k6.s1.lcfalse.extend-c0.18
H7FWB6__real vs B8XC04 (8 other)
region_evalue 3.47, region_ka_bits 33.76, 30 of 49 identical, longest matching h/p run 10

query     131 LTTAVGDMQIAYDNAAGRLNPDFLNLGAGTIGGKTLTPGLYKWTSTLNI 180
              ||||| ||| |    | ||  ||| ||| |   ||  |||||||||
target    144 LTTAVSDMQTAFNDATGRVTPDFTNLGGGELGGLVLTPGLYKWTGAVSV 193

query  h/p    hpphhpphphhhpphhpphphphhphphpppphphphhphpphph
              ||||||||| ||||| |||||||| ||| ||||||  |||||||||| |||
target h/p    hpphhpphphhpphpppphphphpphpppphphhhphphphphphh
```